



COMPLETE MEMORY MAP

For quick revision, remember the lecture in this sequence:

AIR TEMPERATURE

Diurnal → Horizontal → Vertical → Inversion

DIURNAL

Minimum → Sunrise

Maximum → 13–14 h

Clear → Larger range

Cloudy → Smaller range

HORIZONTAL

Isotherm = equal temperature

Equator → Poles = generally decreasing temperature

PBL

Day → Convective mixed layer

Night → Stable boundary + residual layer

INVERSION

Normal: $T \downarrow$ with height

Inversion: $T \uparrow$ with height

Effects → fog + visibility + cloud/precipitation

GDD

$$GDD = T_{\max} + T_{\min} - T_b \quad \boxed{GDD = \frac{T_{\max} + T_{\min}}{2} - T_b}$$

Base temperature = below this plants don't grow

Typical threshold = 4.5–12.5°C; commonly 6°C

HEAT INJURY

Sun scald → high day + low night temperature

Stem griddle → high soil temperature

COLD INJURY

Chilling → low temperature

Freezing → ice crystals

Suffocation → oxygen deficiency

Heaving → ice lifts plant + soil

CROP TEMPERATURE

0–60°C → many higher plants growth range

10–40°C → optimum range for many

20–30°C → maximum matter production

Cardinal temperatures → minimum + optimum + maximum

SOIL TEMPERATURE

Solar radiation + Wind + Evaporation/condensation + Rainfall

•

Aspect/slope + Texture + Tillage + Organic matter + Moisture

SOIL EFFECTS

Temperature → germination + roots + microbes + nutrients + N mineralization + decomposition + weathering + structure

SOIL TEXTURE

Sandy → heats faster

TILLAGE

Loosened soil → larger diurnal range

Night → colder/frost-prone

ORGANIC MATTER

Dark → absorbs more

Organic matter → ↓ thermal conductivity, ↓ heat capacity, ↑ water holding

MOISTURE

Moist soil → evaporation → cooling

Moist soil → better heat conduction → more uniform temperature

This gives you the **full Lecture 3A content in a recall-friendly form**, without bringing in outside material.

Lecture 3A — 35 Important MCQs

These are selected to cover the **maximum concepts, definitions, numerical values, formulas, tables, and likely MCQ traps** from the lecture. **Only the slide content is used.**

1. Minimum air temperature generally occurs at:

- a. Midnight
- b. Sunset
- c. Sunrise
- d. Noon

Answer: c

2. Maximum air temperature is generally recorded between:

- a. 6–8 AM
- b. 10–11 AM
- c. 13–14 hrs
- d. 18–20 hrs

Answer: c

3. The daily range of air temperature is calculated as:

- a. $T_{min} - T_{max}$
- b. $T_{max} - T_{min}$
- c. $T_{max} + T_{min}$
- d. T_{max}/T_{min}

Answer: b

4. Compared with cloudy days, clear days generally have:

- a. Smaller daily temperature range
- b. Larger daily temperature range
- c. No temperature variation
- d. Constant temperature

Answer: b

5. In the Northern Hemisphere, winter minimum temperature generally occurs in:

- a. December
- b. January
- c. June
- d. July

Answer: b

6. Lines connecting points having equal temperature are called:

- a. Isobars
- b. Isohyets
- c. Isotherms
- d. Isotachs

Answer: c

7. In general, temperature decreases horizontally from:

- a. Poles to equator
- b. Equator to poles
- c. Sea to land only
- d. Mountains to valleys

Answer: b

8. During daytime, the Planetary Boundary Layer is mainly characterized by:

- a. Stable boundary layer
- b. Convective mixed layer
- c. Residual layer only
- d. Strongly stable atmosphere throughout

Answer: b

9. During nighttime, the Planetary Boundary Layer consists mainly of:

- a. Convective mixed layer + clouds
- b. Stable boundary layer + residual layer
- c. Mixed layer + entrainment zone only
- d. Free atmosphere only

Answer: b

10. In the daytime surface layer, the lapse rate shown in the slide is:

- a. Strongly stable
- b. Weakly stable
- c. Nearly adiabatic
- d. Super-adiabatic

Answer: d

11. In the daytime mixed layer, the lapse rate is:

- a. Nearly adiabatic
- b. Strongly stable
- c. Super-adiabatic
- d. Weakly stable

Answer: a

12. At night, the stable boundary layer has a:

- a. Super-adiabatic lapse rate
- b. Strongly stable lapse rate
- c. Nearly adiabatic lapse rate
- d. Unstable lapse rate

Answer: b

13. Temperature inversion occurs when temperature:

- a. Decreases with height normally
- b. Increases with height instead of decreasing
- c. Remains constant with height
- d. Becomes zero with height

Answer: b

14. Which of the following is associated with temperature inversion?

- a. Fog formation
- b. Atmospheric visibility changes
- c. Cloud and precipitation effects
- d. All of the above

Answer: d

15. A degree-day combines:

- a. Pressure and humidity
- b. Time and temperature
- c. Rainfall and humidity
- d. Wind and radiation

Answer: b

16. The minimum threshold temperature is the temperature:

- a. Above which plants stop growing
- b. Below which plants do not grow
- c. At which plants always grow optimally
- d. At which plants die immediately

Answer: b

17. The usual range of minimum threshold temperature for different crops given in the slide is:

- a. 0–5°C
- b. 4.5–12.5°C
- c. 15–25°C
- d. 20–30°C

Answer: b

18. The most commonly used threshold temperature mentioned is:

- a. 2°C
- b. 4°C
- c. 6°C
- d. 10°C

Answer: c

19. The correct GDD formula is:

- a. $T_{\max} + T_{\min} - T_b$
- b. $T_{\max} - T_{\min} - 2 - T_b \frac{T_{\max} - T_{\min}}{2} - T_b$
- c. $T_{\max} + T_{\min} - 2 - T_b \frac{T_{\max} + T_{\min}}{2} - T_b$
- d. $T_{\max} + T_b - 2 - T_{\min} \frac{T_{\max} + T_b}{2} - T_{\min}$

Answer: c

20. Which of the following is an application of GDD?

- a. Estimating crop growth stages
- b. Predicting crop maturity
- c. Timing fertilizer/pesticide application
- d. All of the above

Answer: d

21. A major limitation of GDD is that it assumes:

- a. A linear relationship between temperature and growth
- b. No relationship between temperature and growth

- c. Constant rainfall
- d. Constant soil moisture

Answer: a

22. Sun scald is caused by:

- a. Low temperature during both day and night
- b. High temperature during day followed by low temperature at night
- c. Excessive rainfall
- d. High humidity throughout the day

Answer: b

23. Stem griddle is mainly caused by:

- a. Low atmospheric temperature
- b. High soil temperature
- c. High rainfall
- d. Strong wind

Answer: b

24. Stem griddle causes plant death primarily by destroying:

- a. Leaves
- b. Roots
- c. Conductive tissues
- d. Flowers

Answer: c

25. Chilling injury is commonly associated with:

- a. Plants adapted to hot climates exposed to low temperature
- b. Temperate plants exposed to drought
- c. Plants exposed to excessive rainfall
- d. Plants exposed to high humidity only

Answer: a

26. Freezing injury occurs when:

- a. Water evaporates from leaves
- b. Water freezes into ice crystals
- c. Soil becomes dry
- d. Air temperature rises rapidly

Answer: b

27. Suffocation injury occurs mainly because:

- a. Excess oxygen enters the soil
- b. Ice/snow prevents oxygen entry into the soil
- c. Soil becomes excessively warm
- d. Plants receive excessive sunlight

Answer: b

28. Heaving refers to:

- a. Drying of plants due to heat
- b. Lifting of plants along with soil due to ice crystals
- c. Yellowing caused by chilling
- d. Scorching of stem

Answer: b

29. Most higher plants generally grow within which temperature range mentioned in the lecture?

- a. 0–20°C
- b. 0–60°C
- c. 10–40°C
- d. 20–30°C

Answer: b

30. Maximum matter production in many crops occurs approximately between:

- a. 0–10°C
- b. 10–20°C
- c. 20–30°C
- d. 40–50°C

Answer: c

31. The minimum, optimum and maximum temperature limits of crop growth are collectively called:

- a. Thermal limits
- b. Cardinal temperatures
- c. Heat units
- d. Degree days

Answer: b

32. The optimum soil temperature range for soil microbial activity given in the lecture is:

- a. 5–10°C
- b. 10–15°C
- c. 18–30°C
- d. 30–40°C

Answer: c

33. Which soil generally warms more rapidly?

- a. Clay soil
- b. Heavy soil
- c. Sandy soil
- d. Moist soil

Answer: c

34. Tillage generally causes cultivated soil to have:

- a. Smaller diurnal temperature amplitude
- b. Larger diurnal temperature amplitude

- c. No temperature variation
- d. Constant temperature

Answer: b

35. Which statement about soil moisture is correct?

- a. Moist soil heats more than dry soil
- b. Moisture has no effect on soil heat capacity
- c. Evaporation from moist soil cools the soil
- d. Dry soil is always a better conductor of heat

Answer: c

Lecture 3B — Detailed but Easy-to-Recall Summary

Here is a **slightly more detailed revision summary** of the lecture. I'm keeping it **strictly based on the slides**, with emphasis on concepts and facts useful for the exam.

1. Soil Temperature Variation with Depth

Soil temperature changes with **time and depth**.

Diurnal variation

During a day, the soil surface experiences strong heating and cooling.

As depth increases:

Depth ↑ → Temperature amplitude ↓ + Phase lag ↑

- **1 cm:** largest diurnal variation; heats and cools rapidly.
- **5 cm:** smaller variation because heat is conducted downward.
- **20 cm:** much smoother variation and noticeable delay in daily maximum.
- **80 cm:** diurnal variation is almost absent.
- Around **1 m:** daily fluctuations are strongly damped.

This happens because of **thermal inertia, heat capacity, thermal conductivity and thermal diffusion**.

Think:

Surface = fast & variable → Depth = smooth & delayed

2. Day-to-Day and Seasonal Variation

Soil temperature is governed by:

Energy absorbed at surface – Energy lost from soil

Energy input

Mainly:

- Solar radiation

Energy loss

Through:

- Radiation
- Sensible heat
- Latent heat

Soil temperature changes at different time scales:

- Diurnally
- Day-to-day
- Seasonally

Important depth difference

Diurnal signal: disappears relatively quickly with depth.

Seasonal signal: penetrates much deeper.

For seasonal variation:

- **5–10 cm:** largest seasonal amplitude.
- **50 cm:** seasonal cycle remains but is moderate.
- **100 cm:** strongly smoothed and delayed.

Thus, deeper soil provides a **more stable thermal environment** for roots and soil organisms.

3. Importance of Soil Temperature

Soil temperature controls the rate of many:

- Physical processes
- Chemical processes
- Biological processes

Crop establishment

It affects:

- Germination
- Emergence
- Planting date
- Early crop growth
- Optimum germination conditions

Root and water relations

It affects:

- Root growth/activity
- Water uptake
- Nutrient uptake
- Soil-water movement
- Water availability

Soil processes

It influences:

- Weathering
- Soil formation
- Microbial activity
- Decomposition
- Chemical reaction rates

Key relationship

Temperature × Soil moisture × Crop type → rate of biological and physical processes in root zone.

4. Soil Temperature and Agriculture

Crop growth

Cold soil:

→ **slow germination**

As temperature approaches optimum:

→ **germination increases**

Suitable soil temperature is also important for:

- Root growth
- Water absorption
- Nutrient uptake
- Crop growth rate
- Time required to reach maturity

Biological activity

Temperature affects:

- Microbial growth
- Microbial reproduction
- Decomposition

Chemical and water processes

Within a suitable range:

Temperature ↑ → **chemical reaction rate generally** ↑

Temperature also influences:

- Water movement
- Evaporation
- Hydraulic processes
- Nutrient transformations

Management

Soil temperature can help guide:

- Sowing

- Planting

Irrigation, residue and mulching can modify soil thermal conditions.

5. Q_{10} — Temperature Dependence

Q_{10} tells us how a process rate changes for a **10°C increase in temperature**.

$$Q_{10} = \frac{\text{Rate}(T+10)}{\text{Rate}(T)}$$

Therefore:

$$\text{Rate}(T+10) = Q_{10} \times \text{Rate}(T)$$

Example

If:

- $Q_{10} = 2$
- Rate at 20°C = 5 units/h

Then at 30°C:

$$\text{Rate} = 2 \times 5 = 10 \text{ units/h}$$

So the rate doubles **for this particular Q_{10}** .

 Important:

The statement "**reaction rates double for every 10°C rise**" is only a **rule-of-thumb**, not a universal law.

Actual Q_{10} depends on:

- Process
 - Soil moisture
 - Substrate availability
 - Temperature range
-

6. Wind — Basic Concept

Wind is the:

Horizontal movement of air

It is generated primarily by **pressure-gradient forces**.

How is pressure difference created?

Regional temperature contrasts:

→ **pressure differences**

Pressure-gradient force:

→ **drives air from high pressure toward low pressure**

Then:

- **Coriolis effect** deflects moving air.
- **Surface friction** reduces speed and changes direction near the ground.

Simple chain

Temperature contrast → Pressure gradient → Wind

Coriolis + friction → modify wind

7. Seasonal Variation of Wind

Wind changes seasonally because:

- Solar radiation changes.
- Land-sea heating contrasts change.
- Monsoon circulation may reverse/strengthen prevailing winds.
- Boundary-layer stability modifies near-surface wind speed.

Agricultural effects of wind

Wind influences:

- Evapotranspiration
- Crop water demand
- Heat exchange
- Moisture exchange
- CO₂ exchange
- Spray drift
- Lodging
- Disease dispersal

- Field microclimate
- Wind energy

Remember:

Wind speed = how fast air moves

Wind direction = where air comes FROM

8. Wind Speed Measurement

Standard wind-speed measurement is commonly made:

- **10 m above ground**
- Over **open, level terrain**

Instrument:

 **Anemometer**

Observed wind speed is affected by:

- Exposure
- Surface roughness

Therefore, wind reports should specify:

Measurement height + averaging period

Mean wind speed

Daily mean

Average of observations over a complete day.

For scalar mean speed:

Direction is not considered.

Monthly mean

Average of daily wind speeds for all days in that month.

Annual mean

Average of daily values across the year.

Example

Daily speeds:

2, 4, 3, 5, 6 m/s

Mean = $\frac{2+4+3+5+6}{5} = 4.0$ m/s

9. Wind Direction Convention

This is a **very important exam point**.

Meteorological wind direction is reported as the direction:

 **FROM** which the wind originates.

Angle	Direction
0° / 360°	North
90°	East
180°	South
270°	West

Example

180° wind

= southerly wind

= comes **FROM** south

= blows **towards** north

Terms

Windward: side facing incoming wind.

Leeward: sheltered side away from incoming wind.

Prevailing wind: direction occurring most frequently during a specified period.

Instrument: Wind vane.

10. Seasonal Wind Regimes

Pre-monsoon / Summer

Strong surface heating:

→ deeper daytime boundary layer.

Land-sea heating contrast:

→ stronger regional circulation.

Dry and convective conditions:

→ strong gusts.

Agricultural effects:

- Higher evaporative demand
- Higher spray-drift risk

Monsoon / Wet Season

Dominated by:

- Large-scale pressure gradients
- Monsoon circulation

Rain/clouds modify:

- Surface heating
- Boundary-layer mixing

Agricultural effects:

- Evapotranspiration
- Lodging
- Disease dispersal

Winter / Dry Season

Stable boundary layer:

→ suppresses vertical mixing.

Cold stable nights:

→ weak surface winds may occur.

Agricultural effects:

- Frost risk
 - Pollutant dispersion
 - Moisture loss
-

11. Southwest Monsoon

Period

🔥 June–September

Wind

🔥 Southwesterly

Source

Moist winds mainly come from:

- Arabian Sea
- Bay of Bengal

Flow

🔥 Ocean → Land

Result

Moisture-laden winds:

→ widespread and often heavy rainfall over India.

A strong thermal low develops over **northwest India** during late spring/early summer.

Remember:

SW Monsoon = June–September = SW winds = moisture = major rainy season

12. Northeast Monsoon

Period

 **October–December**

Wind

 **Northeasterly**

After SW monsoon withdrawal:

→ pressure increases over northern India.

↓

Continental flow reverses.

↓

NE winds become established.

These winds generally move:

Northeast → South/Southeast

After crossing the **Bay of Bengal**, they can acquire moisture.

This produces rainfall especially over:

- Tamil Nadu
- Coastal Andhra Pradesh
- Adjoining areas

Thus, NE monsoon is particularly important for:

 **Southern and southeastern peninsular India**

13. Why Do Monsoon Winds Reverse?

The fundamental reason is:

🔥 **Differential heating of land and ocean**

Summer

Land heats strongly

↓

Low pressure develops over heated land

↓

Moist air moves Ocean → Land

↓

Summer monsoon rainfall.

Winter

Land cools

↓

High pressure develops over land

↓

Winds move Land → Ocean

Therefore, the monsoon is a **seasonal circulation system** driven by differential heating.

It is modified by:

- Topography
- Large-scale atmospheric circulation

The **Himalayas and Tibetan Plateau** also influence Asian monsoon circulation.

14. SW vs NE Monsoon — Memorize This Table

Feature	Southwest	Northeast
Period	June–September	October–December
Wind	SW	NE
General flow	Ocean → Land	Land → Ocean
Main moisture	Arabian Sea + Bay of Bengal	Bay of Bengal after crossing
Rainfall	Large parts of India	Southern & southeastern India
Pressure	Thermal low over heated land	Higher pressure over cooled land

15. Monsoon & Indian Agriculture

Monsoon is extremely important because a large portion of Indian agriculture depends on rainfall.

The slide states that the humid southwest monsoon brings substantial rainfall from approximately:

 **early June to October**

The amount and distribution of rainfall influence:

- Crop selection
- Cropping season
- Agricultural productivity

Kharif crops

Require:

High temperature + heavy rainfall

Examples:

- Rice
- Jute
- Sugarcane

Rabi crops

Require:

Moderate temperature + rainfall

Examples:

- Wheat
- Barley

Rubber

Requires:

Uniformly high temperature + rainfall throughout the year

Rubber is therefore grown in the **southern parts of the Deccan**, where temperature remains fairly high and rainfall is well distributed over about **6–8 months**.

16. Monsoon Variability

Monsoon rainfall is classified based on:

- Amount
- Distribution
- Time variation

The four categories shown are:

1. Scanty Monsoon

Rainfall:

<75% LPA <75%\ LPA

 **Less than 75% of Long Period Average**

2. Normal Monsoon

Rainfall:

75%–125% LPA 75\%-125%\ LPA

3. Excess Monsoon

Rainfall:

>125% LPA>125%\ LPA

 **More than 125% LPA**

4. Near Normal

The slide separately describes the summer/Southwest rainfall from **March–May** in relation to LPA and places the remaining years not included in the other categories into this category.

17. Historical Frequency

For the historical record **1901–2015**, the slide states approximately:

- **2 years out of 9 → deficient/scanty**
- **2 years out of 9 → excess**
- **5 years out of 9 → normal**

 Memorize:

2 : 2 : 5

Scanty : Excess : Normal

Final Exam Revision Sheet

Soil temperature

Depth ↑ → amplitude ↓ + phase lag ↑

1 cm → largest daily variation

80 cm/~1 m → daily variation strongly damped

Seasonal variation penetrates deeper than diurnal variation.

Soil temperature importance

Germination + roots + water/nutrients + microbes + decomposition + weathering + chemical reactions

Key factors:

Temperature × Moisture × Crop type

Q_{10}

$$Q_{10} = \frac{\text{Rate}(T+10)}{\text{Rate}(T)}$$

10°C increase → rate changes according to Q_{10}

Doubling is **not universal**.

Wind

Wind = horizontal air movement

Pressure-gradient force → High → Low

Coriolis → deflection

Friction → reduces speed + changes direction

Anemometer → wind speed

Wind vane → wind direction

Standard height → 10 m

Wind direction

0/360° → N

90° → E

180° → S

270° → W

Direction is FROM where wind comes.

Monsoon

SW → June–September → Ocean → Land → major rainfall

NE → October–December → Land → Ocean → Bay of Bengal moisture → southern/southeastern rainfall

Cause of reversal

Differential heating of land and ocean

Monsoon classification

<75% LPA → Scanty

75–125% → Normal

>125% → Excess

Historical ratio → 2 : 2 : 5

Agriculture

Kharif → high temperature + heavy rainfall → rice, jute, sugarcane

Rabi → moderate temperature + rainfall → wheat, barley

Rubber → uniformly high temperature + rainfall throughout year

This is the level of detail I'd use for **one final revision before attempting the 30–35 MCQs** from this lecture.

give all possible mcqs that are important . give 30-35

Lecture 3B — 35 Important MCQs

These cover the **maximum important concepts, formulas, numerical values, definitions, comparisons, and likely exam traps** from Lecture 3B. **Only the lecture content is used.**

1. As soil depth increases, the amplitude of diurnal soil temperature variation generally:

- a. Increases
- b. Decreases
- c. Remains constant
- d. Becomes zero immediately

Answer: b

2. As soil depth increases, the phase lag of soil temperature:

- a. Decreases
- b. Remains constant
- c. Increases
- d. Becomes zero

Answer: c

3. Which depth experiences the largest diurnal variation in soil temperature?

- a. 1 cm
- b. 20 cm
- c. 50 cm
- d. 80 cm

Answer: a

4. At approximately 80 cm depth, diurnal soil-temperature variation is:

- a. Maximum
- b. Very large
- c. Almost absent
- d. Equal to surface temperature

Answer: c

5. Which statement correctly compares diurnal and seasonal soil-temperature variations?

- a. Both disappear at the same depth
- b. Diurnal variation penetrates deeper than seasonal variation

- c. Seasonal variation penetrates deeper than diurnal variation
- d. Neither is affected by depth

Answer: c

6. Soil acts as a thermal buffer mainly because of its:

- a. Colour only
- b. Thermal properties such as heat capacity and thermal conductivity
- c. Wind speed only
- d. Rainfall only

Answer: b

7. Which of the following is an important effect of soil temperature on crop establishment?

- a. Germination
- b. Emergence
- c. Early crop growth
- d. All of the above

Answer: d

8. Soil temperature influences all of the following EXCEPT:

- a. Root growth
- b. Water uptake
- c. Nutrient uptake
- d. Wind direction

Answer: d

9. Soil temperature strongly influences:

- a. Microbial activity and decomposition
- b. Only rainfall
- c. Only wind speed
- d. Only atmospheric pressure

Answer: a

10. Q_{10} describes the change in a process rate for a temperature increase of:

- a. 1°C
- b. 5°C
- c. 10°C
- d. 20°C

Answer: c

11. The correct expression for Q_{10} is:

- a. $Q_{10} = \frac{\text{Rate}(T)}{\text{Rate}(T+10)}$
- b. $Q_{10} = \text{Rate}(T+10) - \text{Rate}(T)$
- c. $Q_{10} = \frac{\text{Rate}(T+10)}{\text{Rate}(T)}$
- d. $Q_{10} = \text{Rate}(T) \times \text{Rate}(T+10)$

Answer: c

12. If $Q_{10} = 2$ and the process rate at 20°C is 5 units/h, the rate at 30°C is:

- a. 2.5 units/h
- b. 5 units/h
- c. 7 units/h
- d. 10 units/h

Answer: d

13. The statement "reaction rates double for every 10°C rise" is:

- a. A universal law
- b. Always true for soil processes
- c. A rule-of-thumb, not a universal law
- d. Applicable only to plants

Answer: c

14. Wind is defined as:

- a. Vertical movement of air only
- b. Horizontal movement of air
- c. Movement of water vapour only
- d. Movement of soil particles

Answer: b

15. Wind is generated primarily by:

- a. Soil moisture
- b. Pressure-gradient forces
- c. Cloud formation
- d. Solar radiation alone

Answer: b

16. Air generally moves due to pressure-gradient force from:

- a. Low pressure to high pressure
- b. High pressure to low pressure
- c. Equator to poles only
- d. Poles to equator only

Answer: b

17. Which force deflects moving air?

- a. Gravity
- b. Coriolis effect
- c. Buoyancy only
- d. Radiation

Answer: b

18. Surface friction near the ground generally:

- a. Increases wind speed
- b. Has no effect on wind

- c. Reduces wind speed and changes its direction
- d. Reverses global circulation

Answer: c

19. Standard wind-speed measurement is commonly made at approximately:

- a. 2 m
- b. 5 m
- c. 10 m
- d. 20 m

Answer: c

20. The instrument used to measure wind speed is:

- a. Wind vane
- b. Anemometer
- c. Thermometer
- d. Hygrometer

Answer: b

21. Meteorological wind direction indicates:

- a. The direction toward which wind is moving
- b. The direction from which wind originates
- c. The speed of wind
- d. The vertical direction of wind

Answer: b

22. A wind direction of 180° indicates:

- a. Northerly wind
- b. Easterly wind
- c. Southerly wind
- d. Westerly wind

Answer: c

23. A southerly wind means the wind:

- a. Moves from north to south
- b. Comes from south and moves toward north
- c. Comes from east and moves west
- d. Comes from west and moves east

Answer: b

24. Which instrument is used to measure wind direction?

- a. Anemometer
- b. Wind vane
- c. Thermometer
- d. Pyranometer

Answer: b

25. The side of an object or landscape facing the incoming wind is called:

- a. Leeward
- b. Windward
- c. Downwind
- d. Sheltered zone

Answer: b

26. The sheltered side away from the incoming wind is called:

- a. Windward
- b. Leeward
- c. Upwind
- d. Frontal

Answer: b

27. A prevailing wind is the wind direction that:

- a. Has the highest temperature
- b. Occurs most frequently over a specified period
- c. Has the highest humidity
- d. Always comes from the ocean

Answer: b

28. The Southwest Monsoon generally occurs during:

- a. January–March
- b. March–May
- c. June–September
- d. October–December

Answer: c

29. During the Southwest Monsoon, the dominant winds are:

- a. Northeasterly
- b. Southeasterly
- c. Southwesterly
- d. Northwesterly

Answer: c

30. The general movement of air during the Southwest Monsoon is:

- a. Land → Ocean
- b. Ocean → Land
- c. Land → Land
- d. Ocean → Ocean

Answer: b

31. The Northeast Monsoon generally occurs during:

- a. June–September
- b. July–October
- c. October–December
- d. January–March

Answer: c

32. The main reason for seasonal reversal of monsoon winds is:

- a. Earth's rotation only
- b. Differential heating of land and ocean
- c. Soil texture
- d. Ocean salinity only

Answer: b

33. During summer, strong heating of land generally produces:

- a. High pressure over land and land-to-ocean flow
- b. Low pressure over land and ocean-to-land flow
- c. Low pressure over ocean and land-to-ocean flow
- d. No pressure difference

Answer: b

34. A scanty monsoon is defined as seasonal rainfall:

- a. Less than 50% of LPA
- b. Less than 75% of LPA
- c. 75–125% of LPA
- d. More than 125% of LPA

Answer: b

35. Which combination correctly represents monsoon rainfall classification?

- a. Scanty <75%, Normal 75–125%, Excess >125% of LPA
- b. Scanty <50%, Normal 50–100%, Excess >100% of LPA
- c. Scanty <60%, Normal 60–120%, Excess >120% of LPA
- d. Scanty <80%, Normal 80–120%, Excess >120% of LPA

Answer: a



Must-Memorize for the Exam

- Depth $\uparrow$ $\rightarrow$ amplitude $\downarrow$ + phase lag $\uparrow$
- 1 cm $\rightarrow$ largest diurnal variation
- ~80 cm $\rightarrow$ diurnal variation almost absent
- Seasonal variation penetrates deeper than diurnal
- $Q_{10} \rightarrow 10^\circ\text{C}$ temperature change
- Wind $\rightarrow$ horizontal movement of air
- Pressure-gradient force $\rightarrow$ high pressure $\rightarrow$ low pressure
- Coriolis $\rightarrow$ deflects wind
- Friction $\rightarrow$ reduces wind speed near ground
- Anemometer $\rightarrow$ wind speed
- Wind vane $\rightarrow$ wind direction
- Wind direction $\rightarrow$ FROM where wind comes
- $180^\circ \rightarrow$ South
- SW Monsoon $\rightarrow$ June–September
- NE Monsoon $\rightarrow$ October–December
- SW Monsoon $\rightarrow$ Ocean $\rightarrow$ Land
- NE Monsoon $\rightarrow$ initially Land $\rightarrow$ Ocean
- Monsoon reversal $\rightarrow$ differential heating of land and ocean
- Scanty $\rightarrow$ $<75\%$ LPA
- Normal $\rightarrow$ $75\text{--}125\%$ LPA
- Excess $\rightarrow$ $>125\%$ LPA